PicoCTF 2026 – Undo

Undo

Easy | General Skills | picoCTF 2026

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Description

Can you reverse a series of Linux text transformations to recover the original flag?

In this challenge, we're asked to reverse some text transformations to get the flag

PicoCTF 2026 – Undo Base64 Encoding / Decoding

```
── Step 1 ──
Current flag: KW5xOW45OG43LWZhMDFnQHplMHN
Hint: Base64 encoded the string.
Enter the Linux command to reverse it:
```

When we connect to the server, we're asked which Linux command will Base64 decode a string

PicoCTF 2026 – Undo
Base64 Encoding / Decoding

```
Base64 Character Set

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmn
opqrstuvwxyz0123456789+/
```

Base64 is a system of encoding that transforms data into a string of characters that includes uppercase / lowercase letters, numbers, and a couple of special characters, like ' + ' and ' / '

PicoCTF 2026 – Undo Base64 Encoding / Decoding

```
└─$ echo -n 'cGljb0NURg==' | base64 -d
picoCTF
```

With Linux, we can use the base64 command with the -d argument to decode Base64 strings back into their original contents

PicoCTF 2026 – Undo String Reverse

```
── Step 2 ──
Current flag: )nq9n98n7-fa01g@ze0sfa4eG-gk3
Hint: Reversed the text.
Enter the Linux command to reverse it:
```

The next step in this challenge requires us to reverse a string, i.e, outputting the string contents in reverse order

PicoCTF 2026 – Undo String Reverse

```
└─$ echo 'picoCTF' | rev
FTCocip
```

Reversing a string in Linux is easy with the rev command

PicoCTF 2026 – Undo Replacing Characters

```
── Step 3 ──
Current flag: cvpbPGS(Eriref1at-g3kg-Ge4a
Hint: Replaced underscores with dashes.
Enter the Linux command to reverse it:
```

The next step in the challenge is to replace the dashes ' - ' in the string with underscores ' _ '

PicoCTF 2026 – Undo Replacing Characters

```
┌──(theshyhat㉿hackerfrogs)-[/tmp]
└─$ echo 'picoCTF' | tr 'pico' 'frog'
frogCTF
```

The Linux tr command can be used to replace characters in strings. The first argument to the command are the characters to target, and the second argument is the characters to change the target to

PicoCTF 2026 – Undo
More Replacing Characters

```
── Step 4 ──
Current flag: cvpbPGS(Eriref1at_g3kg_Ge4afs0e
Hint: Replaced curly braces with parentheses.
Enter the Linux command to reverse it:
```

The next step is to replace more characters with the tr command

PicoCTF 2026 – Undo ROT13 Encoding / Decoding

```
── Step 5 ──
Current flag: cvpbPGS{Eriref1at_g3kg_Ge4a
Hint: Applied ROT13 to letters.
Enter the Linux command to reverse it:
```

The last step asks us to apply a ROT13 transformation to the string

PicoCTF 2026 – Undo
ROT13 Encoding / Decoding

```
ROT13 Encryption

A B C D E F G H I J K L M
| | | | | | | | | | | | |
N O P Q R S T U V W X Y Z
```

ROT13 encryption is a common application of the Caesar cipher, a classical encryption method where the letters in a string are shifted a number of letters to form a ciphertext

PicoCTF 2026 – Undo ROT13 Encoding / Decoding

```
└─$ echo 'picoCTF' | tr 'A-Za-z' 'N-ZA-Mn-za-m'
cvpbPGS
```

In Linux, we can also use the tr command to perform a ROT13 transformation, since we are replacing certain letters with others